

# Robot-assisted photodynamic therapy

Grishacheva T.G.<sup>1</sup>, Vasiliev A.S.<sup>2</sup>, Grabovskiy A.V.<sup>3</sup>, Nikitin S.A.<sup>3</sup>, Kharlamov V.V.<sup>3</sup>, Potrahov N.N.<sup>4</sup>,  
Obornev A.D.<sup>5</sup>, Petrishchev N.N.<sup>1</sup>

<sup>1</sup> Laser medicine center, Pavlov First Saint Petersburg State Medical University, St. Petersburg, Russian Federation

<sup>2</sup> LLC Renomed, St. Petersburg, Russian Federation

<sup>3</sup> LLC Medical Robotics, St. Petersburg, Russian Federation

<sup>4</sup> Department of Electronic Instruments and Devices, Saint Petersburg Electrotechnical University «LETI», St. Petersburg, Russian Federation

<sup>5</sup> Department of thoracic surgery, Saint-Petersburg State Research Institute of Phthisiopulmonology of the Ministry of Healthcare of the Russian Federation, St. Petersburg, Russian Federation

**Abstract—** Medical robot assisted fluorescence diagnostics (FD) and photodynamic therapy (PDT) of malignant lesions of external localizations, as well as intraoperative PDT is presented. This technique provides accuracy and uniformity of laser radiation distribution on the object and safety of treatment protocol.

**Keywords—** medical robot, photodynamic therapy,

## I. INTRODUCTION

Photodynamic therapy (PDT) is known as an effective way to treat malignant tumors and many other diseases of non-tumor nature [1]. On average, about 16 thousand operations are performed in Russia per year according to the data on the procurement of photosensitizers [2]. There is a significant growth of research works in the field of photodynamic theranostics.

New strategies for improvig the efficacy and tolerability of PDT are under continuous development. PDT has proven its efficacy but PDT dosimetry does not take into account heterogeneity of geometrically complex irradiation surfaces and uniformity of radiation distribution in the laser beam. As a result, the compliance with the medical protocol can be in question. Robot-assisted system can ensure significant progress in solving these problems.

The robot-assisted system consists of a dual-wavelength laser module (405 and 660 nm); a manipulator for navigation; a video endoscopic visualization system for the irradiated area.

The system is designed for PDT and FD for both external localizations and intraoperative PDT. The device helps the physician to determine the necessary photosensitizer dose, drug-light interval and photoactivation parameters based on telemetry data.

Built-in clinical protocols allow standardization of the method and minimize the impact of human error in PDT.

The robot system provides high-precision navigation of the source of exposure, visualization of the irradiation zone, dosimetric control and generally increases the safety of FD and PDT procedures.

Algorithms of the proposed system provide the possibility of multiposition irradiation taking into account the surface curvature.

The robotized device is currently being tested in animal experiments (fig. 1).

Robot-assisted photodynamic therapy is a high-tech method, which has high positioning accuracy of laser photoactivation compared with traditional PDT, and can reduce manual error and, thus, enable a higher level of safety.



Fig. 1. Setup for robot-assisted fluorescence diagnostics and photodynamic therapy in experiment

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